

Introduction to Statistical Methods
(S2-25 AIMLCLZC418) – Assignment 2

AIML – Section 7

Each question carries 5 marks .

Total - 10 Marks

Duration: 25th July 2026 – 8th August 2026

- 1) Submissions are individual**
- 2) Solve these on paper, scan, and upload**
- 3) Plagiarism results in zero marks**
- 4) Write your name, BITS ID and Section on each page**
- 5) Only handwritten solutions with formula, full steps with proper justification are required.**

Q1. A public health research institute is conducting a wellness assessment to understand the relationship between daily physical activity and resting cardiovascular health among adults enrolled in a community fitness program. Regular physical exercise is widely recognized as an important factor in maintaining cardiovascular fitness. One commonly used physiological indicator of cardiovascular health is the resting heart rate (RHR), measured in beats per minute (bpm). Individuals with better cardiovascular fitness generally tend to exhibit a lower resting heart rate.

To investigate this relationship, the research team randomly selected 11 adults participating in the program. Each participant's average daily exercise duration (in minutes) and resting heart rate (in beats per minute) were recorded after four weeks of continuous participation.

The collected data are presented below.

Participant	Exercise Duration (Minutes) (X)	Resting Heart Rate (bpm) (Y)
1	15	85
2	20	82
3	25	81
4	30	77
5	35	76
6	40	74
7	45	73
8	50	71
9	55	69
10	60	67
11	65	65

The research team wishes to determine whether a linear relationship exists between exercise duration and resting heart rate and to interpret the practical implications of the findings.

- a) As an analyst assisting the research team, organize the given observations into a suitable computational table by generating all additional quantities required for evaluating the linear association between the two variables.
- b) Using the completed computational table, obtain the aggregate statistics required to estimate the strength of the linear relationship between exercise duration and resting heart rate.
- c) Employ an appropriate statistical technique to quantify the degree of linear association between the two variables. Clearly present each computational step and obtain the numerical value of the coefficient.
- d) Critically interpret the computed coefficient with respect to both its direction and magnitude. Explain what the result indicates about the relationship between exercise duration and resting heart rate among the sampled participants.
- e) Based on your statistical findings, prepare a brief recommendation for the wellness program coordinator regarding whether increasing regular exercise duration appears to be associated with improved cardiovascular fitness. Support your conclusion using appropriate statistical evidence.

[5 M]

Q2. A leading national bank has recently launched a customer service improvement initiative aimed at reducing customer waiting times at its urban branches. As part of this initiative, the bank introduced different operational strategies at three branches. These strategies include increasing the number of service counters during peak hours, implementing a token-based queue management system, and providing specialized customer service representatives to handle routine banking transactions.

After implementing these strategies for one month, the bank's Operations Management Team intends to evaluate whether the average customer waiting time differs among the three branches. Since waiting time directly influences customer satisfaction and service quality, identifying any significant differences among the branches will help management determine whether a particular operational strategy is more effective than the others.

To conduct the study, the waiting times (in minutes) of six randomly selected customers were recorded during peak business hours from each branch. The collected data are presented below.

Customer	Branch A	Branch B	Branch C
1	18	15	12
2	20	16	13
3	19	17	14
4	21	18	15
5	22	16	13
6	20	17	14

Assume that customer waiting times are normally distributed and that the population variances are equal. At the 5% level of significance, determine whether there is sufficient statistical evidence to conclude that the average waiting time differs among the three bank branches.

- (a) As a data analyst working for the Operations Management Team, summarize the waiting-time data by determining the sample size, group totals, group means, overall total, and grand mean required for conducting a One-Way Analysis of Variance (ANOVA).
- (b) Using the information obtained in part (a), compute the Correction Factor (CF), the Total Sum of Squares (SST), the Sum of Squares Between Groups (SSB), and the Sum of Squares Within Groups (SSW). Clearly show every intermediate step involved in the calculations.
- (c) Construct the complete One-Way ANOVA table by determining the appropriate degrees of freedom, mean squares, and the F-statistic.
- (d) Using the critical value approach, test whether there is a statistically significant difference in the average customer waiting times among the three branches. Clearly state the null hypothesis, alternative hypothesis, decision rule, and conclusion.
- (e) Based on your statistical findings, prepare a recommendation for the bank's senior management regarding the branch operations. Identify which branch recorded the lowest average waiting time and discuss the practical implications of the results. State that while ANOVA determines whether significant differences exist among the branch means, additional post hoc analysis would be required to identify which specific branch means differ significantly. **[5 M]**
